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Research on the optimization of credit score card for solving the QUBO model based on annealing algorithm and quantum computer

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***Abstract*—The optimization algorithm of the quantum computer can better find the correlation between the different factors in the credit score card, so as to better predict the credit risk. By establishing a multivariate nonlinear integer programming model and transforming it into a QUBO model [1], the bank credit score card is combined optimized to maximize the total bank revenue. First, we establish a multivariate nonlinear integer programming model to convert the quadratic Hamiltonian operator into the form of multiple matrix products. We use the quantum annealing algorithm to find a credit score card in 100 credit score cards and find the corresponding threshold to maximize the total revenue target function. Secondly, on the basis of the above model, we choose the designated three credit score cards and only select one set of thresholds for each credit score card, and find the threshold values corresponding to the three credit score cards, so that the total income target function is the largest. Finally, we arbitrarily selected three credit score cards and only each credit score card chooses one set of threshold, and found the threshold corresponding to the three credit score cards, so that the total income target function is the largest. The above are converted to QUBO form solution [2].**

Keywords—annealing algorithm, quantum computer, QUBO model

I. Introduction

Quantum computers can play an important role in the combination optimization of credit score cards. In credit evaluation, quantum computers can obtain more accurate credit scores by optimizing multiple factors, such as credit history, income, and debt. Through the optimization algorithm of the quantum computer, the correlation between the different factors in the credit score card can be better found, so as to better predict the credit risk. In addition, quantum computers can also obtain more reasonable credit score results by optimizing different weights in the credit score card. Traditional weight calculation methods often need to determine the weight through trial and error, while the quantum computer can quickly find the optimal weight combination through the quantum optimization algorithm.

This paper aims to explore the feasibility and effectiveness of solving the combination optimization problem of credit score card, and compare the difference and advantage with the classical algorithm. By experimenting and testing on real credit score card data sets, we expect to prove the superiority of quantum annealing algorithm [3], [4] in the field of credit score card, provide a new credit evaluation method, and provide more effective theoretical guidance and practical application for risk management.

II. A Multivariate Nonlinear Integer Programming Model

A. Set up the objective function

Based on the expression of the bank's total income in the problem, we maximize the bank's total income and obtain the objective function:

$$\begin{aligned} S &= Mpt(1-h) - Mth \\ &= Mpt - M(p+1)th \\ &= Mp \cdot \sum_{j=1}^{100}\sum_{i=1}^{10} x_{ji}t_{ji} - M(p+1)\cdot\left(\sum_{j=1}^{100}\sum_{i=1}^{10} x_{ji}\cdot t_{ji}\right)\cdot h_{ji} \\ &= Mp \cdot \sum_{j=1}^{100}\sum_{i=1}^{10} x_{ji}t_{ji} - M(p+1)\left(\sum_{j=1}^{100}\sum_{i=1}^{10} x_{j}\cdot t_{ji}\cdot h_{ji}\right) \end{aligned} \tag{1}$$

As a matrix expression:

$$S = XMpT - XM(p+1)\begin{pmatrix} t_{1,1}h_{1,1} & \cdots & t_{100,1}h_{100,1} \\ \vdots & & \\ t_{1,10}h_{1,10} & \cdots & t_{100,10}h_{100,10} \end{pmatrix} \tag{2}$$

Where $X^T = (x_{1,1} \cdots x_{1,10} \cdots x_{100,10})_{1\times 1000}$

$$A = \begin{pmatrix} Mpt_{1,1} - M(p+1)t_{1,1}h_{1,1} & \cdots & Mpt_{100,1} - M(p+1)t_{100,1}h_{100,1} \\ Mpt_{1,2} - M(p+1)t_{1,2}h_{1,2} & \cdots & Mpt_{100,2} - M(p+1)t_{100,2}h_{100,2} \\ \vdots & & \vdots \\ Mpt_{1,9} - M(p+1)t_{1,9}h_{1,9} & \cdots & Mpt_{100,9} - M(p+1)t_{100,9}h_{100,9} \\ Mpt_{1,10} - M(p+1)t_{1,10}h_{1,10} & \cdots & Mpt_{100,10} - M(p+1)t_{100,10}h_{100,10} \end{pmatrix} \tag{3}$$

That is, $S = XA$, where i and j are uniquely determined, namely h_{ji}, is determined by t_{ji}.

Arrange the elements of matrix A in rows into 1×1000 vector L, build the matrix K so that the diagonal element of K is the vector element of vector L, K is the diagonal matrix,

$$K = \begin{pmatrix} k_{1,1} & \cdots & 0 \\ \vdots & & \vdots \\ 0 & \cdots & k_{1000,1000} \end{pmatrix}_{1000\times 1000} \tag{4}$$

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Since x_{ji} is a 0-1 variable, we take $x_{ji} = x_{ji}^2$.

$$S = X^T K X \tag{5}$$

B. Constructing the Hamiltonian operator

We introduce the notion of the Hamiltonian operator. The Hamiltonian operator is an important concept in quantum mechanics used to describe the evolution of physical systems at different points in time. In the optimization problem, the Hamiltonian operator can be applied to the constraints in some specific situations.

Specifically, we can consider representing the constraint as a Hamiltonian operator and adding it to the objective function to obtain the new constrained objective function. This constrained objective function can be used to optimize to satisfy all the constraints and find the optimal solution.

$$H(t) = H_{pot}(t) + H_{kin}(t) \tag{6}$$

In this equation, the quantum Hamiltonian *H(t)* represents the evaluation function of the quantum annealing algorithm, *Hpot(t)* represents the potential energy, corresponding to the evaluation function of the simulated annealing algorithm, *Hkin(t)* represents the kinetic energy, and quantum fluctuations are added to the kinetic energy.

Since each credit score card has and can only choose 1 threshold value, $\sum_{i=1}^{10} x_{ji} = 1$.

Only 1 possibility is available at 100x10 possibilities, $\sum_{i=1}^{10}\sum_{j=1}^{100} x_{ji} = 1$.

So, we construct the Hamiltonian operator with the above S additional penalty function:

$$H = XK - \lambda_1\left(\sum_{i=1}^{10}\sum_{j=1}^{100} x_{ji} - 1\right)^2 \tag{7}$$

Where, $x_{ji}(1 \le j \le 100,\ 1 \le i \le 10,)$ is $a_1, a_2, \ldots a_{1000}$.

$$\begin{aligned} H &= XK - \lambda_1\left(\sum_{i=1}^{10}\sum_{j=1}^{100} x_{ji} - 1\right)^2 \\ &= XK - \lambda_1\left[\sum_{i=1}^{1000} a_i^2 + 2(a_1a_2 + a_1a_3 + \cdots + a_{999}a_{1000}) - 2\sum_{i=1}^{1000} a_i + 1\right] \\ &= XK - \lambda_1\left[\sum_{i=1}^{1000} a_i^2 + 2(a_1a_2 + a_1a_3 + \cdots + a_{999}a_{1000}) - 2\sum_{i=1}^{1000} a_i^2 + 1\right] \\ &= X^T(K - W)X \end{aligned} \tag{8}$$

Where, the matrix W is the:

$$W = \begin{pmatrix} -\lambda_1 & \cdots & \lambda_1 \\ \vdots & \ddots & \vdots \\ \lambda_1 & \cdots & -\lambda_1 \end{pmatrix}_{1000\times1000}$$

C. Quantum annealing algorithm solves the QUBO model

The quantum annealing algorithm is a metaheuristic tool, mainly used to solve binary optimization problems. D-Wave Systems's QPU quantum annealing hardware is specifically designed specifically for complex combinatorial optimization problems [5].

$$Obj(x, Q) = x^T \cdot Q \cdot x \tag{9}$$

Here, x, is a vector of N binary variables, while Q is a real matrix of N×N, used to describe the relationship between the variables. Based on the given matrix Q, the objective function in the minimization formula is equivalent to solving a known NP-Hard problem —— that minimizes the lsing model, that is, to find the optimal binary variable allocation scheme. [6] The quantum annealing process is shown in Fig. 1.

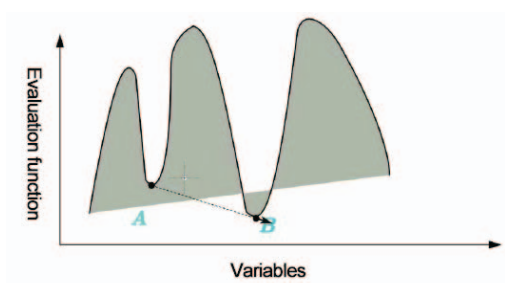

Fig. 1. Quantum annealing process

We define a function named quadratic _ to _ qubo, which takes as input a list of quadratic polynomial coefficients, and returns a corresponding QUBO matrix. Within the function, the number of variables and the size of the QUBO matrix are first determined, and then each element is calculated by a double loop.

1) Algorithmic steps of quantum annealing

a) The Hamiltonian function S (t) is used as the evaluation function.

b) Initialize each parameter: initial temperature $T0$, the initial transverse field strength $\lambda 0$, affects the transition between quantum states; the maximum iteration number Max Steps, initialized state s, corresponding state energy $S(s)$.

c) Random perturbation produces a new state s', corresponding to the state energy $S(s')$. The energy difference $\Delta S = S(s)\text{-}S(s')$, if $\Delta S<0$, the system accepts the new solution $s=s'$, otherwise, the new solution is accepted with the probability of $e^{-\frac{\Delta S}{T}}$.

d) Conduct the cooling operation, $\lambda = \lambda - (\frac{\lambda_0}{MaxSteps})$, if $\lambda = 0$, the quantum annealing algorithm ends, otherwise repeat step c).

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e) By adjusting the initial parameters, the optimal parameter and the corresponding solution under the data of the problem are obtained, and they tend to be stable, and the obtained optimal solution is printed out. The quantum annealing solution process and the difference between the results and the optimal solution are shown in Fig. 2, Fig. 3.

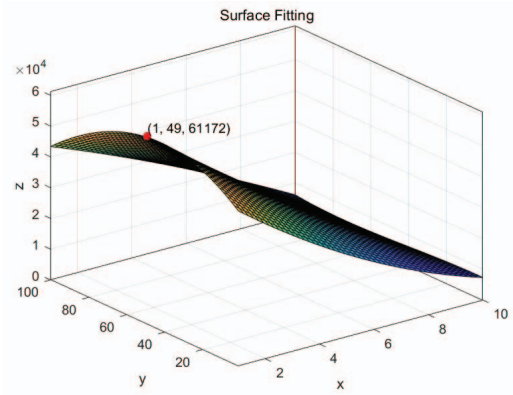

Fig. 2. The quantum-annealing solution process

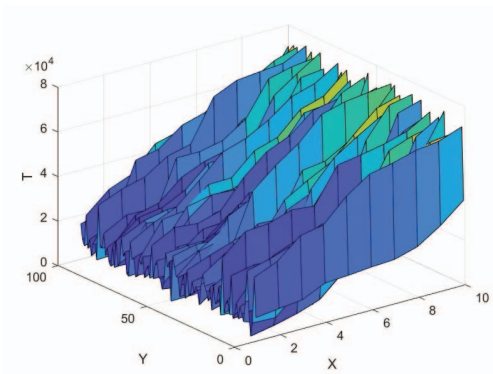

Fig. 3. The difference between the quantum annealing at this time and the optimal solution

2) Model solution

The solution result is: 49 1 61172.0.

That is, only choose Card No. 49 among the 100 credit score cards, and the threshold value is 1, which can maximize the total income of the bank, which is 61172.0 yuan.

III. Specify 3 Credit Score Cards and Seek the Maximum Bank Income

A. Set up the objective function

$$\begin{aligned} H &= MpT(1-H) - MTH \\ &= MpT - M(p+1)TH \\ &= Mp \cdot \prod_{j=1}^{3}\prod_{i=1}^{10} t_{ji} \cdot \left[1 - \frac{M(p+1)}{3}\left(\sum_{j=1}^{3}\sum_{i=1}^{10} x_{ji} \cdot h_{ji}\right)\right] \end{aligned} \tag{10}$$

B. Constructing the Hamiltonian operator

Since each credit score card has and can only choose 1 threshold value, $\sum_{j=1}^{3} x_{ji} = 1$.

Select 3 thresholds out of 30 possibilities for the first 3 credit score cards, $\sum_{i=1}^{10}\sum_{j=1}^{3} x_{ji} = 3$.

So, we construct the Hamiltonian operator with the above S additional penalty function

$$H = S - \lambda_2\left(\sum_{j=1}^{3} x_{ji} - 1\right)^2 - \lambda_3\left(\sum_{i=1}^{10}\sum_{j=1}^{3} x_{ji} - 3\right)^2 \tag{11}$$

C. Simulated annealing to solve the QUBO model

In the quantum annealing algorithm, because the matrix is a tool used to represent the Hamiltonian and calculate the evolution of the quantum state, it searches the evolution of the Hamiltonian to search the minimum energy state, but due to the large order of the matrix, the Hamiltonian is difficult to calculate and measure by matrix. So we use the traditional simulated annealing algorithm to gradually reduce the energy by randomly changing the solution and accepting new solutions with higher probability.

1) Simulated annealing algorithm steps

a) Construct the evaluation function of the temperature system according to the bank's final income formula.

b) Initialize each parameter. Initial temperature T_0, initialized to the temperature throughout the world, cooling coefficient δ, a positive number slightly less than 1, termination temperature T_k, a number slightly greater than 0. The initial state s_0 is the three initial subscripts (x_i, y_i), corresponding to the credit score card and threshold at the position in the table, the initial solution is the initial energy E_0, obtained by the evaluation function under s_0.

c) Take the current state s as a particle, disorder movement, the amplitude of the motion is related to the current temperature T, that is, the higher the temperature, the larger movement range, on the basis of random perturbation to produce a new state s', if s' does not meet the constraint a), b) accept the new state, until meet the constraints, here does not meet the constraint probability is small, acceptable. The new state s' is inserted into the evaluation function to obtain the new solution in the corresponding state, that is, the system energy $E(s')$ in the current state. The difference of the calculation between the current optimal solution and the new solution $\Delta E = E(s) - E(s')$ indicates the energy difference before and after the perturbation. If $\Delta E < 0$, the system accepts the new solution $s = s'$; otherwise, the new solution is accepted with the probability of $e^{-\frac{\Delta S}{T}}$.

d) Conduct the cooling operation, let the current temperature by the cooling coefficient δ to get the cooling temperature T', if $T' > Tk$, repeat the operation c), otherwise, the system is considered to have reached the balance in the low energy ground state.

e) By adjusting the initial parameters, the optimal parameter and the corresponding solution under the data of the problem are obtained, and they tend to be stable, and the obtained optimal solution is printed out. Simulated annealing to solve the optimal combination and simulated annealing combination strategy are shown in Fig. 4, Fig. 5.

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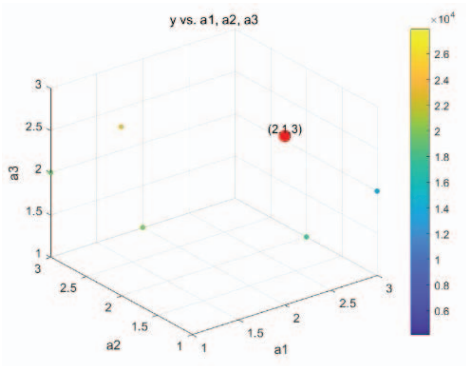

Fig. 4. Simulated annealing to solve the optimal combination

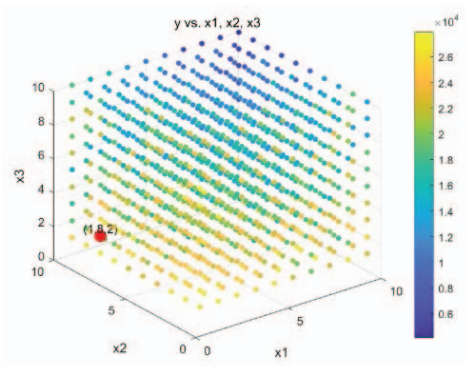

Fig. 5. Simulated annealing combination strategy

2) Solve the results

The eighth threshold of the first credit score card, the first threshold of the second credit score card, and the second threshold of the third credit score card will make the maximum final income at 27914.8. The optimal strategy is shown in Table I.

TABLE I. OPTIMUM STRATEGY

Card number	Threshold group number
1	8
2	1
3	2

IV. CHOOSE ANY 3 CREDIT SCORE CARDS AND SEEK THE MAXIMUM BANK INCOME

A. Set up the objective function

$$\begin{aligned} H &= MpT(1-H) - MTH \\ &= MpT - M(p+1)TH \\ &= Mp \cdot \prod_{j=1}^{100}\prod_{i=1}^{10} t_{ji} \cdot \left[1 - \frac{M(p+1)}{3}\left(\sum_{j=1}^{100}\sum_{i=1}^{10} x_{ji} \cdot h_{ji}\right)\right] \end{aligned} \tag{12}$$

Since x_{ji} is a 0-1 variable, we take $x_{ji} = x_{ji}^2$.

$$Q = \begin{pmatrix} Mpt_{1,1} - M(p+1)\frac{t_{1,1}}{3}h_{1,1} & \cdots & Mpt_{100,1} - M(p+1)\frac{t_{100,1}}{3}h_{100,1} \\ \vdots & & \vdots \\ Mpt_{1,10} - M(p+1)\frac{t_{1,10}}{3}h_{1,10} & \cdots & Mpt_{100,10} - M(p+1)\frac{t_{100,10}}{3}h_{100,10} \end{pmatrix} \tag{13}$$

B. Construct the Hamiltonian operator and the objective function

Since each credit score card has and can only choose 1 threshold value, $\sum_{j=1}^{100} x_{ji} = 1$.

Choose 3 from 100 credit score cards, $\sum_{i=1}^{10}\sum_{j=1}^{1000} x_{ji} = 3$.

So, we construct the Hamiltonian operator with the above S additional penalty function:

$$H = S - \lambda_4\left(\sum_{j=1}^{100} x_{ji} - 1\right)^2 - \lambda_5\left(\sum_{i=1}^{10}\sum_{j=1}^{1000} x_{ji} - 3\right)^2 \tag{14}$$

C. Simulated annealing algorithm was used for solving the QUBO model

1) Simulated annealing algorithm steps

a) Construct the evaluation function of the temperature system according to the bank's final income formula.

b) Initialize each parameter. Initial temperature T_0, initialized to the temperature throughout the world, cooling coefficient δ, a positive number slightly less than 1, termination temperature Tk, a number slightly greater than 0. The initial state s_0 is the three initial subscripts (x_i, y_i), corresponding to the credit score card and threshold at the position in the table, the initial solution is the initial energy E_0, obtained by the evaluation function under s_0.

c) Take the current state s as a particle, disorder movement, the amplitude of the motion is related to the current temperature T, that is, the higher the temperature, the larger movement range, on the basis of random perturbation to produce a new state s', if s' does not meet the constraint a), b) accept the new state, until meet the constraints, here does not meet the constraint probability is small, acceptable. The new state s' is inserted into the evaluation function to obtain the new solution in the corresponding state, that is, the system energy $-E(s')$ in the current state. The difference of the calculation between the current optimal solution and the new solution $\Delta E = E(s) - E(s')$ indicates the energy difference before and after the perturbation. If $\Delta E < 0$, the system accepts the new solution $s = s'$, otherwise, the new solution is accepted with the probability of $e^{-\frac{\Delta S}{T}}$.

d) Conduct the cooling operation, let the current temperature by the cooling coefficient δ to get the cooling temperature T', if $T' > Tk$, then perform operation c), otherwise the system is considered to have reached the balance in the low energy ground state.

e) By adjusting the initial parameters, the optimal parameter and the corresponding solution under the data of the problem are obtained, and they tend to be stable, and the obtained optimal solution is printed out.

2) Model solution

The second threshold of the 8th credit score card, the sixth threshold of the 33rd credit score card and the third threshold of the 49th credit score card will make the maximum final income of 43880.9 yuan. The optimal strategy is shown in Table II.

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TABLE II. OPTIMUM STRATEGY

Card number	Threshold group number
8	2
33	6
49	3

V. RESULTS ANALYSIS AND MODEL TESTING

A. Interpretation of result

For banks, the higher the pass rate, the more loan applicants can be approved. That means the banks can earn more interest income. However, a high pass rate is usually accompanied by a high bad debt rate, and bad debt mean the risk of loss of capital.

The results of II are far greater than the income of III and IV, so the results are true and reasonable.

B. Model testing

In II, the amount of data is small, the complexity of the problem is extremely simple, and the enumeration algorithm is considered, and the code running time is within the acceptable range. Therefore, enumerate each threshold value of each credit score card, calculate the corresponding loan interest income, and then compare it to find the maximum value of the final income. The simulated annealing to solve the optimal combination and the combination strategy are shown in Fig. 6, Fig. 7.

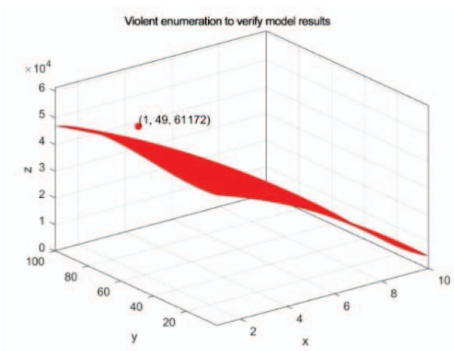

Fig. 6. Simulated annealing to solve the optimal combination

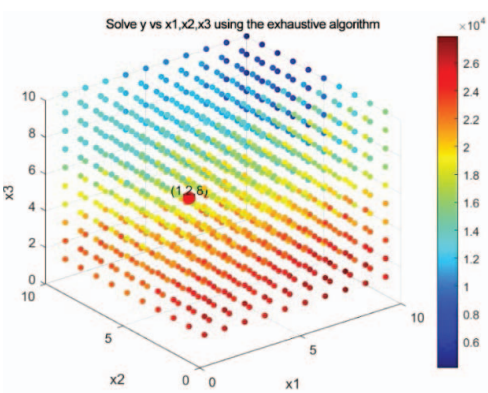

Fig. 7. Simulated annealing combination strategy

In III and IV, the data volume is small and the problem complexity is relatively simple, so the depth-first search algorithm is considered, and the code running time is within the acceptable range.

Therefore, through the depth-first search algorithm and the corresponding constraints, each combination of three different credit score cards and the corresponding threshold is calculated for the corresponding loan interest income. If the result is better, the answer is updated to solve the maximum value of the final income.

After comparing the results, the results are consistent with the strategy selection of the above model establishment, so we believe that the model is more consistent and practical.

VI. EVALUATION OF THE MODEL

A. Advantages of the model

a) The model is suitable for NP-Hard problems and can be effectively solved to find better solutions in a short time relative to other algorithms.

b) The quantum annealing algorithm is used to find the lowest energy state by slowly changing one Hamiltonian to another Hamiltonian, so as to obtain the optimal solution of the combinatorial optimization problem.

c) The model can be extended to solve larger scale optimization problems, and often does not need to redesign the entire system.

B. The shortcomings of the model

a) The annealing algorithm is a probability-based global optimization method with randomness.

b) The accuracy of the QUBO model is limited by the number of binary qubits, and may be more necessary when complex problems need to be solved, leading to a decrease in accuracy.

c) For some complex problems, the solution of the QUBO model may be less easy to find the global optimal solution, but can get only one suboptimal solution

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